

Zekun Wang

Researcher @ Qwen Team

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Research Interests

Model/Data-Efficiency & Acceleration: Efficient architecture for Transformers or hybrid ones, Pruning, distillation, quantization .etc to reduce model size and speedup inference, or efficient training LLMs or MLLMs with a small cost (time or data).

Multi-modal Models and Applications: Large multi-modal models (comprehensive, generation, or unify the both), which support diverse tasks and can be applied as agents in digital or embodied environments.

EXPERIENCE

Alibaba Qwen Team.

Researcher

Beijing, China

Sept. 2025 - Present

- Worked on LLM and Pretraining.

Alibaba Qwen Team.

Research Intern

Beijing, China

May. 2024 - Sept. 2025

- Worked on pre-training and model architectures.

Huawei Noah's ark Lab.

Research Intern

Beijing, China

Jul. 2023 - Apr. 2024

- Worked on improving the efficiency of diffusion models by model compression and step acceleration.

Microsoft Research Asia (MSRA), Natural Language Computing Group.

Research Intern

Beijing, China

Sept. 2020 - Sept. 2021

- Worked with Researcher Wenhui Wang and Dr. Furu Wei on large-scale pre-trained models and efficient methods (like knowledge distillation) in multimodality and natural language processing.

Joint Laboratory of HIT and iFLYTEK Research, Reading Comprehension Group

Research Intern

Beijing, China

June 2019 - Aug. 2019

- Explored the generalization of pre-trained models on different question answering datasets and got the 3rd place in MRQA 2019 shared task@EMNLP 2019.

EDUCATION

Ph.D., Harbin Institute of Technology

Major: Computer Science

Harbin, China

Sept. 2019 - Sept. 2025

B.E., Harbin Institute of Technology

Major: Software Engineering

Harbin, China

Sept. 2015 - June 2019

PUBLICATIONS

* denotes equal contributions, † indicates corresponding author

- Qwen3-Next: Towards Ultimate Training & Inference Efficiency.

Core Contributor

Technical Blog. 2025.

- Qwen3 Technical Report.

Core Contributor

Technical Report. 2025.

- Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free.

Zihan Qiu*, **Zekun Wang***, Bo Zheng*, Zeyu Huang*, Kaiyue Wen, Songlin Yang, Rui Men, Le Yu, Fei

Huang, Suozhi Huang, Dayiheng Liu, Jingren Zhou, Junyang Lin.
NeurIPS 2025 Oral.

- Scaling Computer-Use Grounding via User Interface Decomposition and Synthesis.
Tianbao Xie, Jiaqi Deng, Xiaochuan Li, Junlin Yang, Haoyuan Wu, Jixuan Chen, Wenjing Hu, Xinyuan Wang, Yuhui Xu, **Zekun Wang**, Yiheng Xu, Junli Wang, Doyen Sahoo, Tao Yu, Caiming Xiong.
NeurIPS 2025 Spotlight.
- EffiVLM-Bench: A Comprehensive Benchmark for Evaluating Training-Free Acceleration in Large Visual-Language Models.
Zekun Wang, MingHua Ma, Zexin Wang, Rongchuan Mu, Ming Liu, Bing Qin.
ACL 2025.
- Demons in the Detail: On Implementing Load Balancing Loss for Training Specialized Mixture-of-Expert Models.
Zihan Qiu, Zeyu Huang, Bo Zheng, Kaiyue Wen, **Zekun Wang**, Rui Men, Ivan Titov, Dayiheng Liu, Jingren Zhou, Junyang Lin.
ACL 2025.
- Aguviz: Unified Pure Vision Agents for Autonomous GUI Interaction.
Yiheng Xu*, **Zekun Wang***, Junli Wang*, Dunjie Lu, Tianbao Xie, Amrita Saha, Doyen Sahoo, Tao Yu, Caiming Xiong.
ICML 2025.
- Improved Diffusion-based Generative Model with Better Adversarial Robustness.
Zekun Wang*, Mingyang Yi*, Shuchen Xue, Zhenguo Li, Ming Liu, Bing Qin, Zhi-Ming Ma.
ICLR 2025.
- AgentTrek: Agent Trajectory Synthesis via Guiding Replay with Web Tutorials.
Yiheng Xu, Dunjie Lu, Zhennan Shen, Junli Wang, **Zekun Wang**, Yuchen Mao, Caiming Xiong, Tao Yu.
ICLR 2025 Spotlight.
- CFSP: An Efficient Structured Pruning Framework for LLMs with Coarse-to-Fine Activation Information.
Yuxin Wang*, Minghua Ma*, **Zekun Wang***†, Jingchang Chen, Huiming Fan, Liping Shan, Qing Yang, Dongliang Xu, Ming Liu, Bing Qin.
COLING 2025.
- CodeElo: Benchmarking Competition-level Code Generation of LLMs with Human-comparable Elo Ratings.
Shanghaoran Quan, Jiaxi Yang, Bowen Yu, Bo Zheng, Dayiheng Liu, An Yang, Xuancheng Ren, Bofei Gao, Yibo Miao, Yunlong Feng, **Zekun Wang**, Jian Yang, Zeyu Cui, Yang Fan, Yichang Zhang, Binyuan Hui, Junyang Lin.
Preprint 2025.
- Qwen2.5 Technical Report.
Qwen Team.
Technical Report. 2024.
- CogGPT: Unleashing the Power of Cognitive Dynamics on Large Language Models.
Yaojia Lv, Haojie Pan, **Zekun Wang**, Jiafeng Liang, Yuanxing Liu, Ruiji Fu, Ming Liu, Zhongyuan Wang, Bing Qin.
Findings of the Association for Computational Linguistics: EMNLP 2024.
- Divide-and-Conquer Meets Consensus: Unleashing the Power of Functions in Code Generation.
Jingchang Chen, Hongxuan Tang, Zheng Chu, Qianglong Chen, **Zekun Wang**, Ming Liu, Bing Qin.
NeurIPS 2024 Oral.
- GUIDE: A Guideline-Guided Dataset for Instructional Video Comprehension.
Jiafeng Liang, Shixin Jiang, **Zekun Wang**, Haojie Pan, Zerui Chen, Zheng Chu, Ming Liu, Bing Qin, Ruiji Fu, Zhongyuan Wang.
IJCAI 2024.
- SmartTrim: Adaptive Tokens and Attention Pruning for Efficient Vision-Language Models.

Zekun Wang*, Jingchang Chen*, Wangchunshu Zhou, Haichao Zhu, Jiafeng Liang, Liping Shan, Ming Liu, Dongliang Xu, Qing Yang, Bing Qin.

COLING 2024 Oral.

- MTGER: Multi-view Temporal Graph Enhanced Temporal Reasoning over Time-Involved Document. Zheng Chu, **Zekun Wang**, Jiafeng Liang, Ming Liu, Bing Qin.
Findings of the Association for Computational Linguistics: EMNLP 2023.
- GTR: A Grafting-Then-Reassembling Framework for Dynamic Scene Graph Generation. Jiafeng Liang, Yuxin Wang, **Zekun Wang**, Ming Liu, Ruiji Fu, Zhongyuan Wang, Bing Qin.
IJCAI 2023.
- Distilled Dual-Encoder Model for Vision-Language Understanding. **Zekun Wang**, Wenhui Wang, Haichao Zhu, Ming Liu, Bing Qin, Furu Wei.
EMNLP 2022.
- Less Is More: Domain Adaptation with Lottery Ticket for Reading Comprehension Haichao Zhu, **Zekun Wang**, Heng Zhang, Ming Liu, Sendong Zhao, Bing Qin.
Findings EMNLP 2021.
- Molweni: A Challenge Multiparty Dialogues-based Machine Reading Comprehension Dataset with Discourse Structure. Jiaqi Li, Ming Liu, Min-Yen Kan, Zihao Zheng, **Zekun Wang**, Wenqiang Lei, Ting Liu, Bing Qin.
COLING 2020.
- TAGNet: A Tiny Answer-Guided Network for Conversational Question Generation. **Zekun Wang**, Haichao Zhu, Ming Liu, Bing Qin.
International Journal of Machine Learning and Cybernetics 2023.
- APSam: An Aggregating-then-Pruning Sampler for Question-Conditional Denoising. Jiafeng Liang, Shixin Jiang, Wei Tang, Ning Wang, **Zekun Wang**, Xun Mao, Kai Lv, Ming Liu, Bing Qin
IEEE Transactions on Circuits and Systems for Video Technology 2025.
- Exploring & exploiting high-order graph structure for sparse knowledge graph completion. Tao He, Ming Liu, Yixin Cao, **Zekun Wang**, Zihao Zheng, Zheng Chu, Bing Qin.
Journal of Frontiers of Computer Science 2025.

HONORS & AWARDS

Outstanding PhD Dissertation Nomination

Harbin Institute of Technology

Harbin, China

2018

National Scholarship

Harbin Institute of Technology

Harbin, China

2018

Silver Medal

The ACM/ICPC Asia Contest EC-Final

Shanghai, China

2017

Silver Medal

The ACM-ICPC Asia Regional Contest Shenyang Site

Shenyang, China

2017

Silver Medal

The CCF Collegiate Computer Systems & Programming contest

Fuzhou, China

2017

SERVICES

Reviewer: ICML, ICLR, NeurIPS, ACL, EMNLP, NAACL, ACL Rolling Review, COLING, WSDM, AAAI, IJCAI

SKILLS

Programming Languages

Python, C/C++

Tools and Frameworks

PyTorch, FairSeq, Huggingface Transformers, Pytorch Lightning